Build Multimodal Generative AI Applications ×

Today's goals

Complete any 3 learning items · 3/3
Complete 3 readings · 3/3
Watch a video

Module 1
Foundations of Multimodal AI

Welcome to the Course

- ✓ Video: Course Introduction
  Video • 4 min
- ✓ Reading: Course Overview
  Reading • 3 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- ✓ Helpful Tips for Course Completion
  Ungraded Plugin • 3 min

Introduction to Multimodal AI: Text and Speech Processing

- ✓ Introduction to Multimodal AI
  Video • 8 min
- ✓ Reading: What is Multimodal Generative AI and Why Does It Matter?
  Ungraded Plugin • 5 min
- ✓ Reading: What is Computer Vision?
  Ungraded Plugin • 7 min
- ✓ Text-to-Speech Technologies
  Video • 8 min
- ✓ Speech-to-Text Technologies
  Video • 7 min
- ✓ Reading: Text Processing, Speech Processing, and Text-to-Speech
  Ungraded Plugin • 7 min
- ✓ Lab: Use Mistral and gTTS to Create Your Personal Storyteller
  Ungraded App Item • 30 min
- ✓ Reading: Challenges in Multimodal AI Integration
  Ungraded Plugin • 6 min
- ✓ Lab: Build a Meeting Assistant with Whisper, LangChain, & Gradio
  Ungraded App Item • 45 min
- ✓ Practice Quiz: Introduction to Multimodal AI: Text and Speech Processing
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- ○ **Reading: Summary and Highlights**
  Reading • 2 min
- ○ Cheat Sheet: Foundations of Multimodal AI
  Ungraded Plugin • 5 min
- ○ Graded Quiz: Foundations of Multimodal AI
  Graded Assignment • 21 min

Module 2
Integrating Visual and Video Modalities

Image Generation and Captioning

- ○ Understanding Image Captioning with Meta's Llama
  Video • 7 min
- ○ Reading: Introduction to Text-to-Video and Image-to-Video Technologies
  Ungraded Plugin • 12 min
- ○ Demo: Text-to-Video Generation with OpenAI's Sora
  Video • 7 min
- ○ Lab: DALL.E Image Generation Guide for Beginners
  Ungraded App Item • 20 min
- ○ Reading: Strengths, Limitations, and Practical Applications of Multimodal Vision Models in Real World Scenarios
  Ungraded Plugin • 8 min
- ○ Lab: Build an Image Captioning System with watsonx and IBM's Granite
  Ungraded App Item • 30 min
- ○ Image Generation and Captioning
  Practice Assignment • 10 min

Module Summary and Evaluation

- ○ Reading: Summary and Highlights
  Reading • 3 min
- ○ Cheat Sheet: Integrating Visual and Video Modalities
  Ungraded Plugin • 15 min
- ○ Graded Quiz: Integrating Visual and Video Modalities
  Graded Assignment • 21 min

Module 3
Advanced Multimodal Applications

Build Advanced Multimodal Applications

- ○ Introduction to Multimodal Retrieval-Augmented Generation (MM-RAG)
  Video • 6 min
- ○ Lab: Build a Style Finder Using Multimodal Retrieval and Search
  Ungraded App Item • 45 min
- ○ Multimodal Chatbots and QA Systems
  Video • 7 min
- ○ Lab: Building Your First GenAI-Powered Image-Based Web Application: AI Nutrition Coach
  Ungraded App Item • 30 min
- ○ Build Advanced Multimodal Applications
  Practice Assignment • 15 min

Module Summary and Evaluation

- ○ Summary and Highlights
  Reading • 2 min
- ○ Cheat Sheet: Advanced Multimodal Applications
  Ungraded Plugin • 10 min
- ○ Graded Quiz: Advanced Multimodal Applications
  Graded Assignment • 21 min

Course Wrap-Up

- ○ Course Wrap-up
  Video • 3 min
- ○ Reading: Congratulations and Next Steps
  Reading • 2 min
- ○ Thanks from the Course Team
  Reading • 2 min

Reading: Summary and Highlights

0:11 / 1:43 1x

Congratulations! You have completed this module. At this point, you know that:

Multimodal AI

- Processes and understands multiple types of data simultaneously
- Key components: Text processing, computer vision, speech processing, text-to-speech, and multimodal fusion
- Applications: Virtual assistants, healthcare, education, autonomous vehicles, content creation, and accessibility
- Challenges: Data alignment, modality imbalance, resource demands, interpretability, bias in training data, handling sensitive information
- Future trends: Unified models, edge computing, self-supervised learning, personalization, ethical AI, multimodal LLMs

Text-to-Speech (TTS)

- Converts written text into natural-sounding speech
- Combines linguistic analysis with speech synthesis
- Traditional systems: Separate text analysis, acoustic modeling, and waveform generation
- End-to-end systems: Use VAEs, normalizing flows, and GANs (e.g., VITS)
- Applications: Accessibility, virtual assistants, education, entertainment, healthcare, navigation
- Challenges: Natural prosody, emotional context, multiple speakers, real-time optimization, multilingual support

Speech-to-Text (STT)

- Converts spoken language into written text
- Involves audio feature extraction, sound understanding, word prediction, transcription matching, and refinement
- Applications: Captioning, virtual assistants, medical transcription, language learning, note-taking, meeting and court transcription
- Challenges: Background noise, speaker variability, real-time processing, domain adaptation, low-resource languages, content understanding
- Future trends: Self-supervised learning, multilingual models, contextual understanding, personalization, edge computing

Mark as completed

Like Dislike Report an issue

Go to next item →